



Modelling Financial Markets

Seminar 18: Systemic Risk and Networks — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ ES: Expected Shortfall
- ▣ FRM: Financial Risk Meter
- ▣ GACV: Generalized Approximate Cross-Validation
- ▣ HSBC: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)
- ▣ ING: ING Groep (Internationale Nederlanden Groep) (the Dutch banking group ING)
- ▣ LRMES: Long-Run Marginal Expected Shortfall
- ▣ MES: Marginal Expected Shortfall
- ▣ SG: Société Générale (the French banking group Société Générale)
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States
- ▣ VAR: Vector AutoRegression
- ▣ VIX: CBOE Volatility IndeX

Outline

Introduction

Part A: Pen and Paper

Part B: Data, Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive CoVaR, Δ CoVaR and MES under the Normal distribution step by step
 - ▣ Compute SRISK and the break-even leverage of a bank
 - ▣ Estimate Δ CoVaR and MES on real data with block-bootstrap intervals
 - ▣ Build and read a Diebold–Yilmaz connectedness table; design a reverse stress test
-
- ▣ Background: Lecture 18; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight pen-and-paper exercises (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** one open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Conventions: daily log returns in %; the level α is a tail probability; VaR, CoVaR and MES are positive losses
- The system of a bank is the equal-weighted portfolio of the **other** banks of its region

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (CoVaR under the Normal distribution), A3 (MES, LRMES and SRISK), A5 (a connectedness table), A7 (reverse stress test)
- Part B: B1 (ΔCoVaR of US banks with bootstrap intervals), B3 (MES and break-even leverage), B5 (sensitivity of the connectedness index), B7 (the FRM penalty in one window)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 (model: A5), A8 (model: A7)
- Part B: B2 (European banks; model: B1), B4 (VaR versus ΔCoVaR ; model: B1, B3), B6 (did connectedness rise after 2020?; model: B5), B8 (stress test of a Romanian bank portfolio; model: A7, B3)
- Part C: C1, open question and project seed; C2, critique an AI answer on CoVaR, MES and SRISK (model: A1, A3)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter18_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `scikit-learn`, `matplotlib`
- Series: six US banks, five European banks, Banca Transilvania and BRD; S&P 500, Euro Stoxx 50, BET, VIX; daily prices January 2010 – 18 September 2026
- Prices joined on common trading days first, then log returns; Romanian banks without the days with no trades

Formulas Used Today

- ▣ Quantile regression at level α : $X^{sys} = a_\alpha + b_\alpha X_i$; $\text{CoVaR}_\alpha = -(\hat{a}_\alpha + \hat{b}_\alpha \hat{q}_\alpha(X_i))$, $\Delta\text{CoVaR}_\alpha = \hat{b}_\alpha(\hat{q}_{50\%}(X_i) - \hat{q}_\alpha(X_i))$ [Adrian & Brunnermeier \(2016\)](#)
- ▣ $\text{MES}_{i,\alpha} = -\mathbb{E}[X_i \mid X_m \leq q_\alpha(X_m)]$ [Acharya et al. \(2017\)](#); $\text{LRMES} = 1 - \exp(-18 \text{MES}^2)$ [Acharya, Engle & Richardson \(2012\)](#)
- ▣ $\text{SRISK} = kD - (1 - k)(1 - \text{LRMES})W$; break-even leverage $L^* = 1 + (1 - k)(1 - \text{LRMES})/k$ [Brownlees & Engle \(2017\)](#)
- ▣ Connectedness [Diebold & Yilmaz \(2014\)](#): $F_i = \sum_{j \neq i} \tilde{\theta}_{ij}$, $T_j = \sum_{i \neq j} \tilde{\theta}_{ij}$, net $T_i - F_i$, total $C = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$
- ▣ Reverse stress test: $f^* = -\ell^* \Sigma b / (b^\top \Sigma b)$, $d^* = \ell^* / \sqrt{b^\top \Sigma b}$

A1: CoVaR under the Normal Distribution [Solved]

Task

- Bank and system returns are jointly Normal with zero means: $\sigma_i = 2\%$, $\sigma_{sys} = 1.5\%$, $\rho = 0.6$
- (a) VaR 1% of the bank and of the system
- (b) The distribution of X^{sys} given $X_i = x$
- (c) CoVaR 1% given the bank at its VaR and at its median; ΔCoVaR 1%

Solution

- $z_{1\%} = -2.326$: $\text{VaR}_i = 2 \cdot 2.326 = 4.65\%$,
 $\text{VaR}_{sys} = 3.49\%$
- $X^{sys} \mid X_i = x \sim N(0.45x, (1.5 \cdot 0.8)^2)$: slope
 $\rho\sigma_{sys}/\sigma_i = 0.45$, conditional s.d. 1.20%
- At $x = -4.65$: $\text{CoVaR} = 0.45 \cdot 4.65 + 1.2 \cdot 2.326 = 4.89\%$
- At the median $x = 0$: $1.2 \cdot 2.326 = 2.79\%$;
 $\Delta\text{CoVaR} = 2.09\% = \rho\sigma_{sys} |z_{1\%}|$

A2: A Riskier Bank That Matters Less [Proposed]

Solution discussed in class

Task

- ▣ Bank B has $\sigma_B = 4\%$ (twice the VaR of the bank in A1) but correlation $\rho = 0.3$ with the same system
- ▣ (a) VaR 1%, CoVaR 1% and ΔCoVaR 1% of bank B
- ▣ (b) Which of the two banks would a regulator who looks only at VaR worry about? Which one contributes more?
- ▣ Model: A1

A3: MES, LRMES and SRISK [Solved]

Task

- Bank and market jointly Normal, zero means; market $\sigma_m = 1\%$ per day; bank beta $\beta = 1.3$
- (a) MES 5%
- (b) MES^{2%} (days with a market fall of more than 2%) and LRMES
- (c) SRISK with $W = 100$, $D = 1\,100$, $k = 8\%$; break-even leverage

Solution

- $ES_{5\%}(X_m) = \varphi(-1.645)/0.05 = 2.06\%$;
MES = $1.3 \cdot 2.06 = 2.68\%$
- $\mathbb{E}[-X_m \mid X_m < -2] = \varphi(2)/\Phi(-2) = 2.37$;
MES^{2%} = $1.3 \cdot 2.37 = 3.09\%$
- LRMES = $1 - e^{-18 \cdot 0.0309} = 1 - e^{-0.555} = 42.6\%$
- SRISK = $0.08 \cdot 1\,100 - 0.92 \cdot 0.574 \cdot 100 = 88.00 - 52.80 = 35.20$
- $L = 12$ vs $L^* = 1 + 0.92 \cdot 0.574/0.08 = 7.60$: a shortfall

A4: How Much Does the Prudential Ratio Matter? [Proposed]

Solution discussed in class

Task

- Same bank as A3, but the prudential capital ratio is $k = 5.5\%$
- (a) SRISK and break-even leverage
- (b) By how much must debt fall (at the same W) to remove the shortfall under $k = 8\%$?
- Model: A3

A5: A Connectedness Table [Solved]

Task

- Generalized variance shares (% , rows sum to 100) for three banks

	1	2	3
□ 1	70	20	10
□ 2	25	60	15
□ 3	5	10	85
- (a) From, to and net for each bank; total connectedness
- (b) Pairwise net spillovers and the network

Solution

- From: 30, 40, 15; to: 30, 30, 25
- Net: +0, -10, +10; total $C = 85/3 = 28.3\%$
- Pairwise: $1 \rightarrow 2: 25 - 20 = 5$; $3 \rightarrow 1: 10 - 5 = 5$; $3 \rightarrow 2: 15 - 10 = 5$
- Bank 3 is the only net transmitter; bank 2 is the only net receiver



A6: A Four-Bank Network [Proposed]

Solution discussed in class

Task

- Generalized variance shares (%) for four banks

	1	2	3	4
1	55	25	15	5
2	20	50	20	10
3	10	15	65	10
4	5	5	10	80

- (a) From, to, net and total connectedness
- (b) Draw the network of pairwise net spillovers; which bank would you monitor first?
- Model: A5



A7: Reverse Stress Test with Two Factors [Solved]

Task

- Portfolio loss $L = -b^\top f$, $b = (1.2, 0.8)$; four-week factor covariance $\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix}$ ($\%^2$), $f \sim N(0, \Sigma)$
- (a) The most plausible scenario with a 20% loss
- (b) Its Mahalanobis distance and its probability under the model

Solution

- $\Sigma b = (24, 14.4)$, $b^\top \Sigma b = 40.32$, $\sqrt{\cdot} = 6.35\%$
- $f^* = -20 \cdot (24, 14.4)/40.32 = (-11.9\%, -7.1\%)$
- Check: $-b^\top f^* = 1.2 \cdot 11.9 + 0.8 \cdot 7.1 = 20$
- $d^* = 20/6.35 = 3.15$; $P(L \geq 20) = \Phi(-3.15) = 0.082\%$
- Any other scenario with the same loss is further away, i.e. less likely under the model

A8: When Diversification Across Factors Disappears [Proposed]

Solution discussed in class

Task

- Same b and Σ as A7, target loss 30%
- (a) f^* , d^* and the probability
- (b) Repeat with uncorrelated factors ($\Sigma_{12} = 0$): is the loss more or less plausible? Why?
- Model: A7

B1: ΔCoVaR 5% of US Banks [Solved]

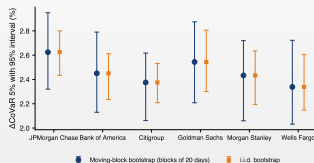
Task

- ▣ Six US banks, 2010–2026; system of bank i = equal-weighted portfolio of the other five [Adrian & Brunnermeier \(2016\)](#)
- ▣ (a) Quantile regression at 5%; VaR 5%, CoVaR 5% and ΔCoVaR 5% of each bank
- ▣ (b) 95% intervals for ΔCoVaR : moving-block bootstrap (blocks of 20 days) [Künsch \(1989\)](#) and i.i.d. bootstrap, 200 resamples each

Interpretation

- ▣ Can the data rank the six banks?
- ▣ Why is the block interval wider than the i.i.d. one?
- ▣ Why do we leave bank i out of its own system?

B1: Results [Solved]



	$\hat{b}_{5\%}$	VaR	CoVaR	ΔCoVaR	block 95% CI	i.i.d. 95% CI
JPMorgan	0.96	2.66	3.81	2.62	[2.32, 2.95]	[2.43, 2.80]
Bank of America	0.76	3.17	3.67	2.45	[2.13, 2.79]	[2.23, 2.61]
Citigroup	0.72	3.25	3.58	2.38	[2.06, 2.62]	[2.21, 2.53]
Goldman Sachs	0.89	2.79	3.93	2.54	[2.21, 2.87]	[2.30, 2.81]
Morgan Stanley	0.72	3.31	3.72	2.43	[2.06, 2.72]	[2.19, 2.64]
Wells Fargo	0.81	2.83	3.89	2.34	[2.03, 2.72]	[2.15, 2.60]

- All six ΔCoVaR lie between 2.34% and 2.62% and every block interval overlaps every other: no reliable ranking
- Block intervals are wider (JPMorgan 0.63 vs 0.37 points): tail days cluster in crises, and the i.i.d. bootstrap breaks the clusters apart
- With bank i inside the system, X_i would appear on both sides of the regression and inflate $\hat{b}$ mechanically

B2: ΔCoVaR of European Banks [Proposed]

Solution discussed in class

Task

- Five European banks; system = the other four
- (a) ΔCoVaR at 1% and at 5%, with a block-bootstrap interval at 1%
- (b) Rank the banks at each level and compare the rankings
- Model: B1

Interpretation

- Does the ranking survive a change of α ?
- How many tail days does each estimate use?
- Would you name one European bank “most systemic” from these data?

B3: MES and the Break-Even Leverage [Solved]

Task

- Thirteen banks; market = S&P 500, Euro Stoxx 50 or BET Acharya et al. (2017)
- (a) MES 5% with a block-bootstrap interval
- (b) $MES^{2\%}$, LRMES and L^* with $k = 8\%$ and $k = 5.5\%$ Acharya, Engle & Richardson (2012), Brownlees & Engle (2017)

Interpretation

- Which region is most exposed to its market?
- How sensitive is L^* to k ?
- What would you need to turn L^* into SRISK in money?

B3: Results [Solved]

	MES 5%	block 95% CI	MES ^{2%}	LRMES	$L^*_{8\%}$	$L^*_{5.5\%}$
JPMorgan	3.22	[2.66, 4.05]	3.61	48%	7.0	10.0
Bank of America	3.85	[3.09, 4.82]	4.38	55%	6.2	8.8
Citigroup	4.03	[3.16, 4.92]	4.58	56%	6.0	8.5
Goldman Sachs	3.34	[2.84, 4.09]	3.78	49%	6.8	9.7
Morgan Stanley	4.04	[3.32, 4.81]	4.62	56%	6.0	8.5
Wells Fargo	3.32	[2.72, 4.07]	3.79	49%	6.8	9.7
Deutsche Bank	4.54	[3.93, 5.12]	4.55	56%	6.1	8.6
BNP Paribas	4.53	[3.83, 5.18]	4.54	56%	6.1	8.6
Santander	4.23	[3.70, 4.77]	4.24	53%	6.4	9.0
ING	4.81	[3.94, 5.67]	4.82	58%	5.8	8.2
HSBC	2.54	[2.08, 2.86]	2.54	37%	8.3	11.9
Banca Transilvania	3.14	[2.61, 3.70]	4.09	52%	6.5	9.2
BRD-Groupe SG	2.69	[2.22, 3.22]	3.53	47%	7.1	10.1

- European banks (except HSBC) have the largest MES; Romanian banks the smallest together with HSBC
- Moving from $k = 8\%$ to 5.5% raises L^* by about 40%: e.g. JPMorgan from 7.0 to 10.0
- SRISK in money needs debt D and market capitalisation W from balance sheets and market data

B4: Does VaR Rank Banks Like ΔCoVaR ? [Proposed]

Solution discussed in class

Task

- Eleven international banks; VaR 1% and ΔCoVaR 1% (system: the other banks of the region)
- (a) Spearman rank correlation between the two measures
- (b) A 95% interval by resampling blocks of days and recomputing both measures for all banks
- Model: B1, B3

Interpretation

- Is the correlation significantly different from zero?
From one?
- What does this imply for bank-by-bank regulation?

B5: How Robust Is the Connectedness Index? [Solved]

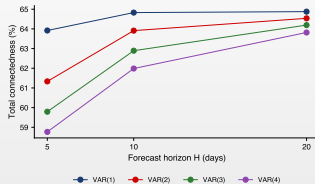
Task

- ▣ VAR on the log Parkinson volatility of 13 banks, 2010–2026 [Diebold & Yilmaz \(2014\)](#)
- ▣ (a) Total connectedness for VAR orders $p = 1, \dots, 4$ and horizons $H = 5, 10, 20$
- ▣ (b) With $p = 1$, $H = 10$: generalized decomposition [Pesaran & Shin \(1998\)](#) vs Cholesky with US banks first and with Romanian banks first

Interpretation

- ▣ Which choice moves the index most?
- ▣ Why does the Cholesky ordering change the net position of US banks?

B5: Results [Solved]



- Generalized index between 58.8% ($p = 4, H = 5$) and 64.9% ($p = 1, H = 20$): stable within about 6 points
- Cholesky gives a lower total (44.1% and 43.5% vs 64.8%): orthogonalisation assigns the common shock to the first variables
- Net position of US banks: +65.9 with US banks first, -95.9 with Romanian banks first, +53.3 generalized
- The ordering, not the data, decides who transmits: use the generalized decomposition

B6: Did Connectedness Rise after 2020? [Proposed]

Solution discussed in class

Task

- Rolling total connectedness (250-day windows, every 5 days) from the lecture
- (a) Mean before and after 1 January 2020
- (b) Regress the index on a post-2020 dummy; t -statistics with ordinary and Newey–West standard errors [Newey & West \(1987\)](#)
- Model: B5

Interpretation

- Why are consecutive values of the index strongly autocorrelated?
- Which t -statistic do you trust?

B7: The FRM Penalty in One Window [Solved]

Task

- JPMorgan regressed at $\tau = 5\%$ on the other 12 banks and on lagged S&P 500, Euro Stoxx 50 and VIX returns, 63 days [Mihoci et al. \(2020\)](#)
- (a) Choose λ by GACV on a grid; count the active coefficients
- (b) A calm window (ending 28 June 2019) and a crisis window (ending 31 March 2020)

Interpretation

- Which banks carry JPMorgan's tail in each window?
- Why is the selected λ so small?

B7: Results [Solved]

- Calm window (2019-03-21 to 28 June 2019): $\hat{\lambda} = 4.5 \times 10^{-5}$, 11 active; largest: Bank of America (0.52), Santander (0.25), Morgan Stanley (0.19), ING (−0.12)
- Crisis window (2019-12-23 to 31 March 2020): $\hat{\lambda} = 7.8 \times 10^{-5}$, 12 active; largest: Bank of America (0.39), Euro Stoxx 50 (t-1) (0.38), BNP Paribas (0.34), HSBC (0.32)
- Bank of America leads in both windows; in the crisis the lagged Euro Stoxx 50 joins the largest coefficients
- 15 regressors for 63 days: the unpenalised fit is feasible, so GACV prefers a small penalty; the FRM needs far more firms than days [Toma et al. \(2026\)](#)

B8: Stress Test of a Romanian Bank Portfolio [Proposed]

Solution discussed in class

Task

- ▣ 1 million RON, half in Banca Transilvania and half in BRD
- ▣ (a) Historical: the worst 10 days in December 2018 – January 2019 and in February – April 2020
- ▣ (b) Hypothetical: weekly betas on Euro Stoxx 50 and BET; apply Euro Stoxx 50 -25% and BET -20%
- ▣ Model: A7, B3

Interpretation

- ▣ Which scenario hurts more: a domestic political shock or a global pandemic?
- ▣ Why is the Euro Stoxx 50 beta close to zero once BET is in the model?

C1: How Exposed Are Romanian Banks to Euro-Area Banking Stress? [Proposed]

Solution discussed in class

- Question: if European banks enter a crisis, how much do Banca Transilvania and BRD lose, and has this exposure grown?
 - ▶ European system: equal-weighted portfolio of Deutsche Bank, BNP Paribas, Santander, ING and HSBC
- Suggested steps
 - ▶ Exposure CoVaR and ΔCoVaR of each Romanian bank given European distress, at 5% and 1%, with block-bootstrap intervals
 - ▶ Compare with the exposure to BET; split the sample 2010–2017 and 2018–2026
 - ▶ Rolling Diebold–Yilmaz: the share of Romanian banks' volatility coming from European and from US banks
- Deliverable: a Quantlet folder and three slides (numbers, interpretation, limitations)

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- ▣ **Student prompt:** "Bank i and the system are jointly Normal with zero means: $\sigma_i = 2.5\%$, $\sigma_{sys} = 1.2\%$, $\rho = 0.5$. Compute CoVaR 1%, ΔCoVaR 1%, MES 5% (the system is the market) and SRISK with $W = 100$, $D = 1100$, $k = 8\%$, LRMES = 40%."
 - ▶ W : market value of equity; D : book value of debt; k : prudential capital ratio
- ▣ **Answer of the AI assistant**
 - ▶ $\text{VaR}_i = 2.5 \cdot 2.326 = 5.82\%$; $X^{sys} \mid X_i = x \sim N(0.24x, 1.04^2)$
 - ▶ CoVaR conditions on the bank at its VaR, $X_i = \text{VaR}_i = 5.82$: the 1% quantile is $0.24 \cdot 5.82 - 1.04 \cdot 2.326 = -1.02$, so CoVaR = 1.02%
 - ▶ Median state: $1.04 \cdot 2.326 = 2.42\%$; $\Delta\text{CoVaR} = 1.02 - 2.42 = -1.40\%$: the bank **reduces** systemic risk
 - ▶ MES is the expected loss of the bank in its own 5% tail: $\text{MES} = 2.5 \cdot \varphi(1.645)/0.05 = 5.16\%$
 - ▶ $\text{SRISK} = kD - (1 - \text{LRMES})W = 88 - 60 = 28$

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

- The answer on the previous slide contains **three** errors typical for systemic-risk measures; the rest is correct
 - ▶ Model: A1 and A3; conventions of this seminar: α is a tail probability, VaR, CoVaR and MES are positive losses
- Your tasks
 - ▶ Find each error: which step, and why it is wrong
 - ▶ Say how you would detect it without knowing the answer: a sign check, a limiting case, a closed form
 - ▶ Give the corrected numbers
 - ▶ Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

Common Pitfalls

- ▣ Putting the bank inside its own system portfolio
- ▣ Ranking banks by ΔCoVaR without intervals, or with i.i.d. intervals that ignore crisis clusters
- ▣ Using Cholesky decompositions whose conclusions depend on the order of the banks
- ▣ Testing a rolling index with ordinary standard errors
- ▣ Reading Normal-model probabilities of reverse stress scenarios as real probabilities

Take-Home Assignment (Team Project)

- ▣ Choose a financial system not covered in class (e.g. Central and Eastern European banks, insurers, crypto exchanges)
- ▣ Estimate MES, ΔCoVaR and a connectedness table, each with block-bootstrap uncertainty
- ▣ Design one historical and one reverse stress test for an equal-weighted portfolio of the system
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ If you use an AI assistant, log every error you catch in [AI_ERRORS.md](#) (as in C2)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Financial Risk Meter for Emerging Markets
- ▣  Dynamic Crypto Networks
- ▣  Network Centrality

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_18

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